

Your Project Objectives (Reframed Clearly)

1. Reduce no-show rate
2. Increase revenue
3. Improve patient satisfaction
4. Utilize doctors effectively across clinics
5. Centralize and organize data

Now we move from **data cleaning** → **modeling** → **KPI creation** → **visualization**.

STEP 1: Confirm Your Data Model (Very Important)

Before calculations, make sure your relationships look like this:

Appointment (Fact Table) connects to:

- Patient (via Patient_ID)
- Provider (via Provider_ID)
- Clinic (via Clinic_ID)
- Date (via Date_ID or Appointment Date)

Appointment should be your central table.

If that's correct, you're ready for KPIs.

STEP 2: Define Core KPIs **(Business-Driven)**

Now let's translate objectives into measurable KPIs.

1 No-Show KPI

Primary KPI:

No-Show Rate %

Formula logic:

No-Show Rate =
 $(\text{Number of No-Show Appointments} \div \text{Total Appointments}) \times 100$

You'll need:

- Appointment Status column
- Identify status = "No-Show"

Also calculate:

- No-show by clinic
- No-show by provider
- No-show by time of day
- No-show by age group
- No-show by day of week

This helps predict patterns.

2 Revenue KPI

Core metrics:

- Total Revenue
- Revenue per Appointment
- Revenue per Provider
- Revenue per Clinic
- Revenue lost due to No-Shows

Revenue Lost =
No-Show Count × Average Revenue per Appointment

That one will impress stakeholders.

3 Doctor Utilization KPI

You mentioned:

Some clinics = 90% utilization

Some = 40%

You need:

Doctor Utilization Rate =
 $(\text{Actual Appointments Attended} \div \text{Available Slots}) \times 100$

If you don't have available slots,
Use proxy:

Appointments per Provider
Appointments per Clinic
Average appointments per day per provider

Then compare clinics side-by-side.

4 Patient Satisfaction KPI

If you have satisfaction scores:

- Average Satisfaction Score
- Satisfaction by Clinic
- Satisfaction vs Wait Time
- Satisfaction vs No-Show

You should definitely analyze:

Wait Time vs Satisfaction

This often shows strong correlation.

5 Wait Time KPI

Since you cleaned it:

- Average Wait Time
 - Wait Time by Clinic
 - Wait Time by Provider
 - % Appointments > 30 minutes wait
-

STEP 3: Recommended Dashboard Pages

Here's how I would structure your Power BI report:

Page 1: Executive Overview

Cards:

- Total Appointments
- No-Show Rate %
- Total Revenue
- Avg Wait Time
- Avg Satisfaction Score

Charts:

- No-show trend over time
- Revenue trend
- Appointments by clinic

Page 2: No-Show Analysis

- No-show by clinic
- No-show by provider
- No-show by age group
- No-show by time of day
- No-show by day of week

This directly supports prediction strategy.

Page 3: Provider Utilization

- Appointments per provider
- Utilization % by clinic
- Comparison: High vs Low performing clinics

This addresses imbalance (90% vs 40%).

Page 4: Patient Experience

- Avg wait time by clinic
- Satisfaction by clinic
- Wait time vs satisfaction scatter plot

This supports experience improvement.

STEP 4: Advanced Insight (If You Want to Impress)

You can create:

- High-Risk No-Show Profile
 - Age group most likely to no-show
 - Time of day highest risk
 - Clinic highest risk
 - Wait time correlation

This helps leadership anticipate no-shows.

Which was one of your original goals.

What You Should Do Next (Practical Step)

1. Create basic measures:
 - Total Appointments
 - No-Show Count
 - No-Show Rate
 - Total Revenue
 - Avg Wait Time
2. Validate numbers.
3. Then move to segmented visuals.

Don't jump to fancy visuals before validating KPIs.

Important Mindset Shift

Your job is not to build charts.

Your job is to answer:

- Why are no-shows happening?
- Where is revenue being lost?

- Which clinics are inefficient?
- What operational change should management make?

Dashboards are just the tool.

Insight is the goal.

Now tell me:

What columns exist in your Appointment table?

Especially:

- Status values
- Revenue column name
- Wait time column name

Let's define your first 3 DAX measures together.

what formula can i use to calculate No-show by clinic and how do i go about it No-show by provider No-show by time of day No-show by age group No-show by day of week

Perfect 🎉 this is where your analysis becomes real.

First, very important:

👉 You do **NOT** create separate formulas for each breakdown.

You create **one No-Show measure**, then slice it by clinic, provider, age group, etc.

Power BI handles the segmentation automatically.



Step 1: Create Base Measures

Assuming:

- Your table is called **Appointment**
 - Status column is **Status**
 - No-show value is **"No-Show"**
-

1 Total Appointments

Total Appointments = COUNTROWS(Appointment)

2 No-Show Count

No-Show Count =
CALCULATE(
COUNTROWS(Appointment),
Appointment[Status] = "No-Show"
)

3 No-Show Rate %

No-Show Rate % =
DIVIDE(
[No-Show Count],
[Total Appointments],
0
)

Format this as Percentage.

Now Here's The Important Part

You do NOT write:

- No-show by clinic formula

- No-show by provider formula
- No-show by age group formula

You use the SAME measure and change the visual.



How To Do Each Breakdown



1 No-Show by Clinic

Create:

- Bar Chart
- Axis → Clinic Name
- Values → No-Show Count OR No-Show Rate %

Power BI automatically filters by clinic because of relationships.

No extra formula needed.



2 No-Show by Provider

- Axis → Provider Name
- Values → No-Show Rate %

Same measure.

Power BI filters by provider context.

3 No-Show by Time of Day

If you have Appointment Time:

First create Hour column:

Hour = HOUR(Appointment[Appointment Time])

Then:

- Axis → Hour
- Values → No-Show Rate %

This shows peak no-show hours.

Very powerful insight.

4 No-Show by Age Group

- Axis → Patient[Age Group]
- Values → No-Show Rate %

Make sure relationship exists between Appointment and Patient.

5 No-Show by Day of Week

If you have Date table:

Create:

Day of Week = FORMAT('Date'[Date], "dddd")

Then:

- Axis → Day of Week
- Values → No-Show Rate %

This helps identify risky days.

Why This Works

DAX works with something called:

Context.

When you put Clinic on the visual:

Power BI automatically filters the Appointment table to that clinic before calculating.

That's why you don't need separate formulas.

If You Want To Be More Advanced

You can create a specific measure like:

```
No-Show Rate by Clinic =  
CALCULATE(  
    [No-Show Rate %],  
    ALLEXCEPT(Appointment, Appointment[Clinic_ID])  
)
```

But honestly, you don't need this for now.

Keep it simple.

What You Just Learned

You moved from:

“How do I calculate this separately?”

To understanding:

“One measure can answer multiple business questions.”

That’s a huge shift.

Your Immediate Action Plan

1. Create the 3 base measures.

2. Build 5 visuals:

- By clinic
- By provider
- By hour
- By age group
- By day

Then tell me:

- Which segment has the highest no-show rate?

That’s your first real insight.